Contest 1 Experiment Report

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Experimental Setup

The labeled dataset contains 3156 rows. The text is drawn from restaurant reviews and includes target attributes we want to predict such as aspect and sentiment. In practice, we want the model to first predict which aspects the text content falls into (is the review talking about the food, service, etc.). Then, we want the model to predict the sentiment towards each aspect of the review. In technical terms, we want the model to sequentially perform multi-label classification to give us aspects of the review then proceed with multi-class classification to pair sentiment label with each one.

We first split the dataset into 80:20 proportions by ids while stratified by the rarity of each aspect & polarity to maintain the same distribution of classes in both training-evaluation set. The training set has 2519 rows. The development set has 637 rows and is used as a hold-out set to evaluate model performance. After some experiment, we've found that most model underperform on classes with low count such as "price", "ambience", and difficult to grasp sentiment like "conflicting" so we use oversampling to increase training samples using duplication & backtranslation (English -> German -> English) to paraphrase the reviews.

The final label distribution is as follows:

```
--- Class Distribution Check ---
Train Rows: 2519
Train (Enriched) Rows: 4103
Dev Rows:   637

--- Aspect Categories ---
food                      | Train: 33.2% (837) | Train (Enriched): 22.7% (931) | Dev: 33.6% (214)
price                     | Train: 8.7% (220) | Train (Enriched): 20.7% (851) | Dev: 8.6% (55)
service                   | Train: 15.9% (401) | Train (Enriched): 10.7% (440) | Dev: 16.5% (105)
ambience                  | Train: 11.8% (297) | Train (Enriched): 21.2% (870) | Dev: 11.1% (71)
anecdotes/miscellaneous   | Train: 30.3% (764) | Train (Enriched): 24.6% (1011) | Dev: 30.1% (192)

--- Polarity Classes ---
negative                  | Train: 23.1% (581) | Train (Enriched): 22.5% (922) | Dev: 21.0% (134)
positive                  | Train: 58.6% (1477) | Train (Enriched): 53.3% (2188) | Dev: 62.6% (399)
conflict                  | Train: 5.6% (141) | Train (Enriched): 8.1% (334) | Dev: 4.1% (26)
neutral                   | Train: 12.7% (320) | Train (Enriched): 16.1% (659) | Dev: 12.2% (78)
```

Aspect	training set	training set (enriched)	dev set
food	837	931	214
price	220	851	55
service	401	440	105
ambience	297	870	71
anecdotes/miscellaneous	764	1011	192

Sentiment	training set	training set (enriched)	dev set
positive	1477	2188	399
negative	581	922	134
neutral	320	659	78
conflicting	141	334	26

Model

As for baseline models, we chose to compare with a logistic regression classifier (w/ TF-IDF) and neural networks utilizing a frozen GloVe 6B (100d) word embeddings as a reliable performance floor as follows: 1. **Deep Averaging Network (DAN):** Averaged word embeddings passed through a feed-forward network 2. **Convolutional Neural Network (CNN):** Utilized 1D convolutions (kernel sizes 3, 4, 5) to capture local n-gram contexts**.**

We hypothesized that utilizing transfer learning by fine-tuning transformer models (through Hugging Face **transformer** API) can help achieve significantly higher accuracy while using a reasonable amount of time & resources to implement, so we experimented with 3 popular pre-trained BERT-based models to find the best fit for our case.

The BERT-based models implemented using Hugging Face API are as follows:

1. **DistilBERT** (distilbert-base-uncased): a distilled, faster version of BERT as an easy-to-use variant.

   Hyperparameters - Batch Size: 16, Learning Rate: 2e-5, Epochs: 3, Max Sequence Length: 128

2. **RoBERTa** (roberta-base): Optimized BERT variant with dynamic masking and trained on more data.

   Hyperparameters - Batch Size: 16, Learning Rate: 2e-5, Epochs: 3

3. **DeBERTa** (microsoft/deberta-base): Utilizes disentangled attention mechanisms to better capture the relative positions of words (crucial for targeted sentiment).

   Hyperparameters - Batch Size: 8, Learning Rate: 2e-5, Epochs: 10 (with **Early Stopping**, patience=2).

Hyperparameters are predefined. We use Adam optimizer and set the learning rates to 2e-5. The minibatch sizes are 8 for DeBERTa and 16 for the other 2 models. For CNN, we use basic settings, three filter sizes (3,4, and 5) at 100 filters. For DAN, we use a Dropout layer to mitigate overfitting at the hidden layer level. The word embeddings (GloVe) are pre-trained with 6B tokens, 100 dim downloaded from the version provided by Stanford then frozen to be used in creating feature vectors as is.

Results

According to the evaluation, all models performed aspect detection better than sentiment analysis giving higher f1 score

Initial evaluations showed that DistilBERT (epochs=2) struggled with **minority classes**, achieving an overall F1 of only 0.5559 and critical error in capturing Price and Conflict. Through error analysis, we observed that the model frequently missed obvious, distinct keywords for underrepresented classes. To correct this, we've found that utilizing **oversampling** on imbalance data & rule-based (common words distinctly associated with those aspects) **heuristic override** approach to help force aspect detection help the model generalize onto lower-count & harder to detect classes. For example, if the word 'cost' or 'expense' appeared we forced the 'price' aspect or the presence of 'noisy' or 'music' forces the 'ambience' aspect in post even If the model doesn't label it as such. These methods result in higher f1 scores as seen in the performance comparison for DistilBERT below:

	Price Aspect F1	Ambience Aspect F1	Conflict Sent F1	Overall F1
DistilBERT **(epochs=2, no oversampling, pre-enrichment)**	0	0.5765	0	0.5559
DistilBERT **(epochs=1, + oversampling & heuristic override)**	0.7730	0.5739	0.0625	0.5813
DistilBERT **(epochs=3, optimal epochs for training)**	0.8429	0.7328	0.4444	0.6923

Our approach of transfer learning using pre-trained BERT models we hypothesized earlier works very well. Even the smallest transformer (**DistilBERT**) significantly outperformed the GloVe-based neural networks across the board, validating attention & BERT ability to capture deep contextual embeddings. Then by scaling up to a bigger model such as **DeBERTa** and finishing up the training phase with **early stoppage** and **threshold tuning** (instead of default cutoff being 0.5 we use a validation split in the

training samples to find the optimal threshold for each aspect that the model will use to decide if we want to keep that label) we achieve an ideal model for this dataset with an **Overall f1 score of 0.7742**

	Aspect Detection F1	Sentiment Analysis F1	Overall F1 (Micro Avg.)
Logistic Regression (TF-IDF)	0.6827	0.6219	0.4603
CNN + GloVe	0.8270	0.7034	0.6149
DAN + GloVe	0.8009	0.7033	0.6073
DistilBERT (epochs=2, no oversampling)	0.7474	0.7105	0.5559
DistilBERT (epochs=1, + oversampling)	0.7922	0.6932	0.5813
DistilBERT (epochs=3, + heuristic override)	0.8638	0.7773	0.6923
RoBERTa (epochs=1)	0.8351	0.7845	0.6811
RoBERTa (epochs=3, optimized)	0.8857	0.8393	0.7630
DeBERTA (early stoppage, optimized)	**0.8895**	**0.8446**	**0.7722**

Conclusion

From our experiment results, we find that **DeBERTa** optimized via **threshold selection** & **heuristic override** to force aspects performs the best for both sentiment analysis and aspect classification tasks. Therefore, we **retrain** this model on the whole labeled dataset then use it to classify the unlabeled contest1_test.csv set and submit to MCV for the ABSA Contest.